MIL-C-688E
6 August 1969
SUPERSEDING
MIL-C-688D
14 June 1965

MILITARY SPECIFICATION

CREAM, STERILIZED

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 Scope.- This specification covers sterilized whipping grade cream for use in the kitchens of the Armed Forces as an item of general issue on a limited basis (see 6.1).

2. APPLICABLE DOCUMENTS

* 2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

TT-C-495	-	Coatings, Exterior, for Tinned Food Cans.
PPP-B-636	-	Box, Fiberboard.
PPP-C-96	-	Cans, Metal, 28 Gage and Lighter.
PPP-F-320	-	Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.
PPP-G-460	-	Glass Containers, One Gallon Capacity and Smaller, For Other Than Medicinal Products, Packaging and Packing Of.

MILITARY

MIL-L-1497	-	Labeling of Metal Cans for Subsistence Items.

FSC 8910

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-668 - Sanitary Standards for Food Plants.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal, shall apply:

U.S. DEPARTMENT OF HEALTH, EDUCATION AND WELFARE

Federal Food, Drug and Cosmetic Act and Regulations Promulgated Thereunder

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D. C. 20402.)

AMERICAN PUBLIC HEALTH ASSOCIATION

Standard Methods for the Examination of Dairy Products

(Application for copies should be addressed to the American Public Health Association, 1790 Broadway, New York, N. Y. 10019.)

U.S. DEPARTMENT OF AGRICULTURE

Working Reference of Livestock Regulatory Establishments, Circuits and Officials

Visual Aids for Inspection of Metal Containers

(Application for copies should be addressed to the U.S. Department of Agriculture, Consumer and Marketing Services, Washington, D. C. 20250.)

ASSOCIATION OF OFFICIAL ANALYTICAL CHEMISTS

Official Methods of Analysis of the Association of Official Agricultural Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, Box 540, Benjamin Franklin Station, Washington, D.C. 20044.)

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UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Classification Committee Agent, Room 202, Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

AMERICAN TRUCKING ASSOCIATION, INC.

National Motor Freight Classification Rules

(Application for copies should be addressed to the American Trucking Association, Inc., 1616 P Street, N.W., Washington, D.C. 20036.)

3. REQUIREMENTS

* 3.1 First article.- When specified (see 6.1) a sample of the finished product shall be submitted to the contracting officer for approval before production is commenced.

* 3.2 Material.- The material components shall comply with table I.

TABLE I.- Components

Item No.	Requirement
1	Raw milk shall be obtained from cows in herds accredited as tuberculosis-free and certified brucellosis-free by the U.S. Department of Agriculture, or herds that have passed an annual tuberculosis test and meet U.S.D.A. requirements for an individually certified herd, or from cows in herds located in (1) a Modified Accredited Tuberculosis Area; and (2) either (a) a Certified Brucellosis-free Area, or (b) a Modified Certified Brucellosis Area; or (3) an area in the process of being accredited or certified by the U.S.D.A. In addition, the milk shall be normal in appearance, practically free from colostrum and have a clean, sweet odor. It shall be subject to inspection by the procuring agency or duly authorized representative. The bacterial estimate at the time of processing, shall not exceed 300,000, per milliliter (ml.) when determined by the standard plate count.
2	Cream shall be plant separated from fresh raw milk meeting the requirements of item 1, and the acidity of the cream, calculated as lactic acid, shall not exceed 0.12 percent.

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TABLE I.- Components (cont'd)

Item No.	Requirement
3	Skim milk, concentrated skim milk or concentrated whole milk if required for standardizing purpose shall be prepared from milk meeting the requirements of item 1, U.S.D.A. Extra grade or better, low heat, non-fat dry milk may also be used for standardizing purposes.
4	Colloidal stabilizers, if used, shall be carrageen, alginate or other suitable vegetable gum.
5	Stabilizing salts, if used, shall be disodium phosphate or sodium citrate.

* 3.3 Processing.- Processing shall be in accordance with table II.

TABLE II.- Processing

Item No.	Requirement
1	All raw milk, unless processed within two hours, shall be cooled immediately after receipt to 45°F. or lower and maintained at that temperature until processing is started.
2	All ingredients shall be heat treated in accordance with a high-temperature-short-time method and shall be aseptically filled and sealed.
3	Steam infusion sterilization methods may be employed provided the system employs steam filtration systems and use of culinary steam in accordance with the recommendations of the Grade "A" Pasteurized Milk Ordinance.

* 3.4 Finished product.- The finished product shall comply with requirements of table III.

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TABLE III.- Finished product

Item No.	Requirement
1	Samples of the canned finished product shall be incubated in accordance with 4.5.1.1. Commercial sterility shall be indicated by absence of swells, springers or leakers, a natural white or light cream color: a sweet flavor and no abnormal flavors such as sour, putrid, bitter or fruity. 1/
2	The product may exhibit a thick, semi-solid, non pourable, consistency, requiring removal from the container by means of a spoon or spatula.
3	There shall be no free or churned fat, and no more than a trace of curd settling or whey separation.
4	The milk fat shall be not less than 30.0 percent.
5	The milk solids not-fat shall be not less than 5.5 percent.

1/ A slight cooked (mildly heated) flavor shall be acceptable.

* 3.5 Whipped product.- The whipped product shall comply with table IV.

TABLE IV.- Whipped product

Item No.	Requirement
1	The cream after whipping as described in 4.5.2.3 shall have obtained an overrun of not less than 100 percent within 4 minutes when tested as specified in 4.5.2.3.
2	The whipped product shall have a natural white or light cream color, a stiff body and smooth texture.
3	When tested for serum separation as described in 4.5.2.4, there shall be not more than 1 ml. of serum drainage.

* 3.6 Sanitary requirements.- Receiving stations, plants, equipment, operations, personnel and transportation facilities shall meet the sanitary standards as defined in MIL-STD-668.

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* 3.7 Storage temperature.- The finished product shall be cooled to a temperature of from 35° to 50°F. within 96 hours after the packaging operation. Temperature for additional storage and to the point of delivery shall be within this range.

* 3.8 All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Pre-acceptance inspection.- Product shall be rejected if produced in plants not meeting the requirements of MIL-STD-668.

4.3 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

* 4.3.1 Component and material inspection.- In accordance with 4.1, components and materials shall be inspected in accordance with the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

* 4.3.1.1 Inspection of milk and milk products.-

* 4.3.1.1.1 Animal sources.- Inspection shall be made to determine that cows supplying the milk used in the preparation of this product are in herds designated by the U.S. Department of Agriculture as tuberculosis-free and from herds designated brucellosis-free or in the process of being so designated. Determination of herd status regarding tuberculosis and brucellosis shall be made by contacting the nearest Office of the Animal Disease Eradication Division of the Consumer and Marketing Service. (Location of this office can be ascertained by consulting the publication entitled, "Working Reference of Livestock Regulatory Establishments, Circuits,

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and Officials".) Nonconformance to either of the above requirements shall be cause for rejection of the lot.

* 4.3.1.1.2 Establishment sources.- Examination shall be made to determine that all processed milk products have been manufactured, packaged and shipped by establishments approved as Military sources of supply as listed in the Directory of Sanitarily Approved Food Establishments for Armed Forces Procurement, or from plants listed with a rating of 90 or more in the document entitled "Sanitation Compliance and Enforcement Ratings of Interstate Milk Shippers". Nonconformance to the above requirements shall be cause for rejection of the lot or the involved quantity of finished product made therefrom, as applicable.

* 4.3.1.2 Examination of raw whole milk.- Examination and testing of raw whole milk shall be made to determine compliance with the requirements of table I, item 1, as concerns appearance, odor and bacterial estimate at the approximate time of use. Records of all required examinations and tests shall be maintained and made available for review. Failure to comply with one or more or the above referenced requirement(s), reflected by actual examination and tests or by examination of records shall be cause for rejection of the involved lot of finished product.

4.3.1.3 Examination of cream, skim milk, concentrated skim milk, concentrated whole milk and nonfat dry milk.- Examination of these products shall be made to determine compliance with the requirements of items 2 and 3 of table I as concerns source, identity and grade, as applicable. Such compliance shall be ascertained by examination of plant records, U.S.D.A. grade certificate or other valid documents, as applicable. Failure to comply with one or more of the above referenced requirements, reflected by actual examination or by examination of records, shall be cause for rejection of the involved lot of finished product.

4.3.1.4 Examination of colloidal stabilizers and stabilizing salts (as applicable).- Examination of colloidal stabilizers and stabilizing salts shall be made to determine compliance with the requirements of items 4 and 5, table I, as concerns identity. Such compliance shall be ascertained by examination of labels, invoices, or other valid documents. Non-compliance with identity requirements shall be cause for rejection of the finished product made therefrom.

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* 4.3.2 Testing of components.- Testing of components described in this specification shall be as shown in table V. Results shall be applicable to the lot average. Nonconformance to one or more test requirements shall be cause for rejection of the lot.

TABLE V.- Testing of components 1/

Component	Sample unit	Lot size expressed	Inspection level	Characteristics	Results reported	Test ref & rqmts. para.
Can and lids 2/	1 can and lid	Cans	S-1	Tin plate	Nearest 0.01 lb/ base box	5.1.1.1 and 4.5.2.5
Metal cap 2/	1 metal cap	Caps	S-1	Tin plate	Nearest 0.01 lb/ base box	5.1.1.2 and 4.5.2.5
Cream (Plant separated)	1-pint	Quarts	3/	Acidity (calculated as lactic acid)	Nearest 0.01 percent	3.2 and 4.5.2.2

1/ Each component or component characteristic, as applicable, which is not listed in this table unless otherwise accepted, shall be tested in accordance with the applicable procedure contained in pertinent subsidiary specifications.

2/ Testing of can and bottle cap coating (when required) and labeling shall be accomplished on the same cans and bottle closures used for testing of tin plate. One or more failures for can coating or labeling shall be cause for rejection of the lot.

3/ The sample for testing shall be an approximate 4-ounce composite derived from 5 primary containers or all containers if less than 5 form a lot.

* 4.3.3 Process examination.- Examination shall be made during processing to determine compliance with requirements contained in table II pertaining to equipment, procedures and controls. Records shall be maintained of time intervals and temperatures. Noncompliance as determined by actual examination or as reflected by the records shall be cause for rejection of the involved finished product.

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* 4.3.4 Examination of end item.- Classification of defects found during examination shall be in accordance with tables VII through XI and 4.3.4.1 through 4.3.4.5. The examination criteria for the above tables shall be as shown in table VI. The acceptable quality levels (AQLs) shall be expressed as percent defective for tables IX and X and in terms of defects per hundred units for tables VII, VIII and XI.

TABLE VI.- End item examination criteria

Table	Inspection level	Sample unit	Lot size expressed in	AQLs Major A	Major B	Minor	Total
VII	I	1 primary container	Primary containers	0.25	1.5	-	4.0
VIII	S-2	1 primary container	Primary containers	-	1.5	-	10.0
IX	S-3	1 can	Cans	-	1.5	-	-
X	S-3	1 primary container	Primary containers	-	-	1.5	-
XI	S-3	1 primary container	Primary containers	-	1.0	-	-

TABLE VII.- External examination of primary containers

Category			Defect	Applicable to Cans 1/	Bottles w/caps
Major A	Major B	Minor			
101			Leaker	X	-
102			Broken or cracked.	-	X
103			False seam or otherwise improperly closed.	X	X

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TABLE VII.- External examination of primary containers (cont'd)

Category			Defect	Applicable to Cans 1/	Applicable to Bottles w/caps
Major A	Major B	Minor			
	151		Not type or size specified.	X	X
	152		Chipped or thin spot.	-	X
	153		Severe dent or dent causing sharp ridge.	X	-
	154		Severe paneling, buckling or collapsed can.	X	-
105			Pitted rust. 2/	X	X
	155		Nomenclature missing, incorrect or illegible.	(see 4.3.4.1)	X
		201	Not clean. 3/	X	X
		202	Moderate dent.	X	-
		203	Other required label information missing, incorrect or illegible.	X	X

1/ Use U.S.D.A. - CMS Visual Aids for Inspection of Metal Containers as a guide for classifying can defects.

2/ Rust that can be removed by wiping with a soft cloth shall not be considered as pitted rust.

3/ Cans showing a very thin film of grease which is discernible to the touch but not readily discernible visually shall be considered to be clean.

TABLE VIII.- Examination for interior coating (applicable to cans and metal bottle closures)

Category		Defect
Major B	Minor	
151		Interior rusted.
152		Enamel missing.
153		Blistered or softened areas which can be peeled by fingertip abrasion.
	201	Blistered or softened areas which cannot be peeled by fingertip abrasion.
154		Scratches through enamel. 1/
155		Denuded areas (other than scratches).

1/ Scratches or absence of enamel resulting from normal fabrication and handling or from opening the sealed can for inspection shall not be considered a defect.

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TABLE IX.- Examination for vacuum (can only)

Category	Defect
Major	
151	Vacuum less than 5 inches. (See 4.5.1.2 for procedure).

TABLE X.- Examination for volume 1/ 2/

Category	Defect
Minor	
201	Volume of contents more than 2 percent under specified volume.

1/ Temperature of contents of filled container shall be 68°F. ± 5°F. at time of examination.

2/ Lot shall be rejected if sample data indicate lot average volume is less than specified volume.

TABLE XI.- Examination for product characteristics and sterility 1/ 2/

Category	Defect
Major B	Unwhipped product 3/
151	Not a pleasant flavor.
152	Not a natural white or light cream color.
153	Presence of churned or free fat.
154	More than a trace of curd settling or whey separation.

1/ Examination shall be accomplished after incubation. See 4.5.1.1 for procedures.

2/ The finding of one or more of the following defects shall be cause for rejection of the lot:

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a/ Presence of foreign material, e.g., wood, metal, glass, insect or insect fragments, dirt, hair, etc.

b/ Presence of abnormal flavors or odors, e.g., sour, putrid, bitter or fruity.

c/ Presence of swells, leakers or springers, as applicable.

3/ No defect shall be scored if body exhibits a semi-solid or non-pourable consistency.

* 4.3.4.1 Examination of can labeling.- Examination of can labeling shall be in accordance with the examination criteria of MIL-L-1497.

* 4.3.4.2 Examination of can coating or bottle closure coating.- When can coating is required, examination shall be in accordance with examination criteria of TT-C-495.

* 4.3.4.3 Examination of shipping containers for cans.- Examination of shipping containers for level B packing shall be in accordance with the examination criteria in the appendix of PPP-B-636. In addition, the following defects shall be included in the table of examination: Major: Marking missing, incorrect or illegible. Minor: Arrangement not as specified. Containers with round wire strapping intended for transfer at sea shall not be acceptable. For level C pack, the aforementioned examination criteria shall be applicable except only the defects pertaining to marking and arrangement shall apply.

* 4.3.4.4 Examination of shipping containers for glass bottles.- Examination of shipping containers for glass bottles for levels B or C packing shall be in accordance with the examination criteria of PPP-G-460, except that for level C packing only the defects for workmanship, content and marking shall apply.

* 4.3.4.5 Examination for storage and destination temperature requirements of finished product.- Examination for storage and destination temperature of the finished product shall be made in accordance with the requirements of 3.7. Daily temperature records shall be maintained. Failure to comply with the temperature requirements reflected by actual examination or by records shall be cause for rejection of the involved lot of finished product.

* 4.3.5 Sampling procedure and acceptance criteria for testing of finished product.- The product shall be tested for milk fat and milk solids-not-fat, as specified in table III, items 4 and 5 and for the whipping test (see 4.5.2.3, item 1, table IV), and serum separation requirements of table IV, item 3.

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Samples submitted to the laboratory for the whipping test shall be stored and shipped in conformance with the range of temperature specified in 3.7. Procedures for testing shall be in accordance with 4.5. The sample unit for testing shall be an approximate 1 quart composite derived from the number of primary containers indicated by inspection level S-1. The composite shall be made just prior to testing. Lot size shall be expressed in terms of primary containers. Results shall be reported as follows: Milk fat and milk solids-not-fat to the nearest 0.1 percent; whipping test to the nearest one percent overrun and serum separation to the nearest 0.5 ml. Nonconformance to one or more test requirements shall be cause for rejection of the involved lot of finished product.

* 4.3.5.1 Examination of the whipped product.- At the time of testing for the specified overrun (see 4.5.2.3), examination of the whipped product shall be made for body, texture and color in accordance with the requirements of item 2 of table IV. Nonconformance to one or more of the aforementioned referenced requirements shall be cause for rejection of the involved lot of finished product.

* 4.4 Standby test samples.- The Government reserves the right to withdraw and withhold standby test samples of components or finished product or both (quantity of which shall be not more than twice that required by this specification) for inspection purposes. Samples not used will be returned to the supplier.

4.5 Test methods.- Examination and test procedures which differ from those specified herein unless otherwise excepted, may be used by the supplier if they provide a quality assurance equivalent to that specified. If the Government inspection activity determines that such procedures and controls do not provide, as a minimum, such quality assurance, the supplier will use the test procedures set forth herein. In case of dispute as to test results, the test methods specified herein will govern.

4.5.1 Examination procedures.-

4.5.1.1 Sterility incubation.- Samples of the filled and sealed primary containers shall be incubated at a temperature of from 90° to 95°F. for 7 days and then examined for compliance with item 1 of table III.

* 4.5.1.2 Can vacuum.- After processing, sample cans shall be selected which are free from dents or other visible damage. The cans and contents shall be allowed to reach a temperature of 68° ± 5°F. The vacuum reading shall then be taken with a puncture-type vacuum gauge making the puncture as near as possible to the double seam to minimize error due to distortion of the end. For normal inspection purposes, a correction of 1 inch of vacuum shall be added to the gauge reading for each 1,000 feet above sea level at which the determination is made. The following is an example of a normal inspection vacuum determination:

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	Inches
Vacuum reading	6
Altitude correction (2,000 ft.)	2
Corrected vacuum	8

If greater accuracy is considered necessary, sample cans shall be submitted to a laboratory for vacuum determination corrected to standard conditions (68°F. and 29.9 inches of mercury). Corrections for gauge and headspace errors shall be included.

4.5.2 Test procedures.-

4.5.2.1 Bacteriological analysis.- Analysis for bacteriological estimate shall be made in accordance with Standard Methods for the Examination of Dairy Products, method: Direct microscopic count.

4.5.2.2 Chemical analyses.- Chemical anaylses of the finished product shall be made in accordance with the following methods from Official Methods of Analysis of the Association of Official Agricultural Chemists, chapter: Dairy Products:

Test	Section	Methods
Raw cream		
Acidity (calculated as lactic acid)	Milk	Acidity (weigh samples)
Finished product		
Milk fat -	Cream	Roese - Gottlieb
Milk solids-not-fat		Total Solids 1/

1/ Determine by subtracting percent of milk fat from total solids.

* 4.5.2.3 Whipping test.- The whipping test shall be conducted as follows:

1. Cream temperature shall be adjusted to from 40° to 45°F. 24 hours before whipping.

2. A Hobart model 4C, or equivalent, mixer equipped with all purpose beater shall be used.

3. One pint of cream at 40° to 45°F. shall be added to a clean, chilled (40°F. or lower), Pyrex or stainless steel bowl.

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4. Start whipping at dial setting No. 2 to avoid splashing. Shift to speed No. 6. When a noticeable volume increase has been obtained, finish whipping at speed No. 10 until standing peaks are formed.

5. A minimum overrun of 100 percent shall be obtained within 4 minutes whipping time when calculated by the following formula:

$$\text{Percent overrun} = \frac{\text{wt. of cream} - \text{wt. of same volume of whipped cream}}{\text{wt. of same volume of whipped cream}} \times 100$$

4.5.2.4 Serum separation test.- Fill an 8 ounce kitchen measuring cup to its capacity with the whipped product. Using a spatula, transfer the contents of the cup onto a 6 inch square of No. 26 B&S gauge 16 mesh woven wire. Place the screen, containing the cream in a single heap, over an appropriate size funnel supported by a "C" ring stand and collect any drainage of serum in a graduate that may be read to 0.5 ml. After two hours exposure to room temperature (70° to 80°F.) measure the quantity of serum drainage. (See item 3, table IV).

4.5.2.5 Tin coating weights.- Tin coating weights shall be determined by any method specified in PPP-C-96.

5. PREPARATION FOR DELIVERY

5.1 Packaging.- The product shall be filled and sealed under aseptic conditions in accordance with level B or C, as specified (see 6.1). The container, lids, and closures shall be sterilized immediately prior to the filling and sealing operation and shall be capable of resisting the effects of sterilization.

* 5.1.1 Level B.- The product shall be packaged in accordance with 5.1.1.1 or 5.1.1.2, as specified (see 6.1).

* 5.1.1.1 Metal cans.- The product shall be filled into cans, the sizes and net contents of which shall be in accordance with table XIII, as specified (see 6.1). Cans shall be round, open-top style, with soldered side seam and compound-lined, double-seamed ends. After filling, the cans shall be hermetically sealed under conditions which will result in a vacuum of not less than 5 inches, when tested in accordance with 4.5.1.2. Cans shall be made from not less than 0.25-pound per base box electrolytic tin place (see 4.5.2.5) and shall be coated overall inside with an enamel furnished by the can manufacturer in accordance with best commercial practice for the product. The cans shall be clean and free from foreign material, flippers, swells, dents, holes, paneling and rust. Cans shall be coated outside with a coating conforming to type I of TT-C-495.

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TABLE XIII.- Can sizes, net contents, number per case, and arrangement in shipping container

Can size	Common nomenclature	Net contents (fl. oz. at 68°F.)	No. per case	Arrangement		
				Length	Width	Depth
307 by 710	-	32	12	4	3	1
401 by 509	-	32	24	4	3	2
404 by 414	No. 3	32	24	4	3	2

* 5.1.1.2 Glass bottles.- The product shall be packaged in 1-quart bottles. The closure shall be a press-on metal cap made from not less than commercial 0.25-pound per base box electrolytic tin plate (see 4.5.2.5), and shall be coated overall inside with an enamel suitable for the product. Bottle closures shall be coated outside in accordance with type I of TT-C-495.

5.1.2 Level C.- Commercial aseptic packaging in metal cans or glass bottles will be acceptable. Can and bottle closures with or without commercial exterior coating will be acceptable.

5.2 Packing.- The product shall be packed in accordance with 5.2.1 or 5.2.2, as applicable.

5.2.1 Cans.- The filled cans, arranged as specified in table XIII, shall be packed in accordance with level B or C, as specified (see 6.1).

* 5.2.1.1 Level B.- The filled cans shall be packed into a fiberboard box, constructed and closed in accordance with style RSC, type CF or SF, class domestic of PPP-B-636. Alternatively, a style RSC, end-opening box shall be used.

* 5.2.1.1.1 When specified (see 6.1 and 6.3) the filled cans shall be packed into a fiberboard box constructed and closed in accordance with style RSC, V3c or V3s of PPP-B-636. Alternatively, a style RSC, end-opening box shall be used. Alternatively, the box may be constructed of fiberboard complying with V4s of PPP-F-320. For transfer at sea the shipping container shall be reinforced with other than round wire strapping in accordance with the appendix of PPP-B-636.

5.2.1.2 Level C.- The filled cans shall be packed in a manner that will insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Shipping container shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

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* 5.2.2 Glass bottles levels B and C.- As specified (see 6.1), the product in glass bottles shall be packed in accordance with the level B or C requirements of PPP-G-460, as specified (see 6.1).

5.3 Labeling and marking.-

* 5.3.1 Cans.- Cans shall be labeled in accordance with MIL-L-1497. Labeling shall include instructions for use.

5.3.2 Glass bottles.- Suitable commercial labeling will be acceptable. Labeling shall include instructions for use.

5.3.3 Shipping containers.- Shipping containers (for cans and bottles), shall be marked in accordance with MIL-STD-129, and as specified in 5.3.3.1. In addition, containers for glass bottles shall be marked in accordance with 5.3.3.2.

* 5.3.3.1 Additional marking.- The following additional precautionary marking shall be printed or stenciled on top of the shipping container in bold letters from 1 to 1-1/2 inches high, all letters being of the same height:

KEEP REFRIGERATED

(35° to 50°F)

5.3.3.2 Containers for glass bottles.- Any special markings specified in PPP-G-460 shall appear on shipping containers containing glass bottles.

6. NOTES

* 6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) First article.- When a preproduction sample is not required (see 3.1).
(c) Levels of packaging and packing required (see 5.1 and 5.2).
(d) Type of packaging required (see 5.1.1).
(e) Size of packaging required (see 5.1.1.1 and 5.1.1.2).
(f) When packing specified in 5.2.1.1.1 is required.

* 6.2 Packing specified in 5.2.1.1.1 is intended to provide limited military protection for specific operational requirements.

6.3 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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U. S. Army Natick Laboratories
Natick, Massachusetts 01760

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DEPARTMENT OF THE ARMY

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Commanding General
U. S. Army Natick Laboratories
ATTN: AMXRES-EQS
Natick, Massachusetts 01760

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

Cream, Sterilized MIL-C-668E

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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